

AWI-ESM3 LR: estimated CMIP7 output volume

TCo95 + FESOM 2.7 core3 + REcoM + LPJ-GUESS | scaled from the measured HR test year cli119 | 14 September 2026

Result

With the HR variable set, LR physics and land produce roughly **41 GB per year**. REcoM adds about **10 GB**, giving a baseline of **about 50 GB per year**, roughly a ninth of HR. Fields that are switched off for HR, and daily 3D biogeochemistry, can raise this to at most about 180 GB per year.

≈ 41 GB	+ 10 GB	≈ 50 GB	5–6.5 TB	≤ 180 GB
physics and land	REcoM, monthly	baseline per year	baseline per 100 years	per year, all options

Physics and land: scaled from HR

HR's cmorized volume was measured per grid on one full model year (529 files) and divided by the ratio of cells, for 3D ocean fields also by the ratio of vertical levels. Sizes are compressed on-disk sizes.

grid	HR per year	HR cells → LR cells	LR per year
native atmosphere and land	274 GB	TCo319 421 120 → TCo95 40 320 (÷10.4)	26.3 GB
regrided atmosphere	68 GB	1296×640 → 400×192 (÷10.8)	6.3 GB
FESOM nodes, 2D	56 GB	DARS 3 146 761 → core3 220 509 (÷14.3)	3.9 GB
FESOM nodes, 3D	54 GB	plus 57 → 48 levels (÷17.0)	3.2 GB
FESOM elements, 2D and 3D	18 GB	6 229 020 → 426 403 (÷14.6 / ÷17.4)	1.1 GB
fixed, decadal, climatology	7 GB	written once per run or decade	<1 GB
total	470 GB		≈ 41 GB

Coarse fields compress slightly worse per value and the fixed overhead per file does not shrink, so 40–55 GB is the realistic range. The file count stays at about 500 per year.

Ocean biogeochemistry: measured from a REcoM test run

HR has no REcoM, so this part comes from RECOM_concdriven_04 (model year 1900), which already runs on the LR ocean. It writes 74 FESOM streams that HR does not have, 13.2 GB per year, all monthly: 53 in 3D (12.4 GB) and 21 in 2D (0.8 GB). Four of them (bolus_u/v/w, std_dens_DIVbolus, 1.9 GB) are eddy-parameterisation diagnostics rather than biogeochemistry, which leaves **about 11 GB of REcoM output**. For monthly 3D ocean fields, cmorizing keeps 82–94 % of the raw size (measured on HR temp and salt), so REcoM adds **about 10 GB per year** if its output maps onto CMIP7 variables as written.

The CMIP7 data request lists 178 ocean biogeochemistry entries (106 variables): 151 monthly and 27 daily. Seven of the daily ones are 3D (detoc, dissoc, phycalc, phypico, zmeso, zmicro, zooc). REcoM currently writes nothing daily; producing those seven would need daily 3D output and would add about **50 GB per year**, about 7.6 GB per field.

Options that raise the total

if produced for LR	added per year
6-hourly on 7 pressure levels (ta, ua, va, hus, zg), fixed in the LR XIOS configuration	+4 GB
3-hourly on 6 pressure levels (ta, ua, va, hus, wap)	+8 GB
6-hourly on model levels (ta, ua, va, hus, zg), 91 or 137 levels	+30 to 50 GB
daily 3D biogeochemistry (seven fields above)	+50 GB
eddy-induced transport (msftmpa, GM is on in LR), ocean tendencies, monthly land and snow fields	< 3 GB

Pressure- and model-level estimates use HR's measured compressed bytes per stored value: about 1 byte for model-level and 1.7 to 2.7 bytes for pressure-level fields.

Not covered

- **Compute time.** HR needs 33 node-hours per model year, but that is dominated by I/O on the large grids and does not scale reliably to LR.
- **Run lengths.** All numbers are per model year; multiply by the length of each LR experiment.
- **Mapping of REcoM streams.** Not all 74 streams become CMIP7 variables one to one; some requested variables are derived (integrals, sums over plankton groups). The estimate assumes the volume stays similar.

Sources: HR /scratch/a/a270092/pycmor_hr/cli119; LR mesh /work/ab0246/a270092/input/fesom2/core3; LR atmosphere and REcoM /work/bb1469/a270092/runtime/awiesm3-develop-cc/ (cc_tco95_c, RECOM_concdriven_04)